

# Tech Assignment: Optimize Product Placement for Earnings

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## Problem Statement

At **Forbes Advisor**, we **drive users to partner websites** (banks, insurance companies, credit cards, etc.) those users then **convert** on partner sites.

Your goal is to help us determine **how to improve the ordering and positioning of products** on our pages to **maximize Earnings Per Visitor (EPV)**. EPV is one of our most important metrics and directly impacts revenue.

### We want to know:

Based on product performance and user behavior, which products should appear in which positions on a page to maximize EPV? Your recommendations should help us answer this:

*“If we’re showing 5 products on a page, which product should go in Position 1, 2, 3, etc. — to maximize total EPV for the page?”*

## Dataset Context

The dataset provided is a **page-level session export**, sourced from:

- **Google Analytics (GA)** for user behavior (e.g., sessions, bounce rate, engagement time)

Each row corresponds to a **single user session** on a product comparison or review page. The data captures **what users did on the Forbes site**, and whether they later **converted on a partner’s site**.

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## Your Target Metric: EPV (Earnings Per Visitor)

We’d like you to **optimize EPV**, calculated as:

$$\text{EPV} = \text{revenue} / \text{sessions}$$

You may also explore it at a **per-product**, **per-page**, or **per-segment** level (e.g., traffic source, device, geography).

## Important Context: Product Position Bias

In real-world user behavior, **products ranked higher on the page** (i.e. closer to the top) receive more attention, more clicks, and more conversions — *regardless of how strong the product actually is*. This phenomenon is known as **position bias**.

This bias makes it hard to separate product quality from visibility. We'd love to see:

- How you would **identify or simulate product position**
- Your thoughts on **how position bias could distort EPV measurements**
- Any ideas for **correcting or modeling around it**

Please share your **code (Jupyter Notebook, Python script, or Colab link)** along with a brief **slide deck or write-up** explaining your approach, key insights, and recommendations. We're looking for clear structure, business understanding, and thoughtfulness.

### DataSet

Attached